M&AE 4070/5070 - Dynamics of Flight Vehicles

Chapter 5 - Longitudinal Dynamics

Companion Notes for Case Study 6

1 Linearization of the longitudinal motion equations

The analysis of an aircraft's dynamics aims at determining how "pilotable" it is, the most elementary aspect being its stability. This naturally leads to the notion of modes associated with a *linear* model of the aircraft. The linearization allows access to the notion of modes and dynamic stability, and serves as basis to develop auto-pilot/feedback controls to improve flying quality.

- **First step: State-space representation.**

$$X = \begin{pmatrix} V \\ \gamma \\ \alpha \\ q \\ h \end{pmatrix} \quad U = \begin{pmatrix} \delta_x \\ \delta_m \end{pmatrix} \quad \frac{dX}{dt} = \dot{X} = f(X,U)$$

$n = 5$ $\quad m = 2$ $\quad$ non linear, vector function

- **Second step: Linearization.** AROUND GIVEN, KNOWN EQUILIBRIUM X_e, U_e

↳ look at $\overline{X} = X - X_e = \Delta X$

$\overline{U} = U - U_e = \Delta U$

Goal : get system of equations of the form $\frac{d\overline{X}}{dt} = A\overline{X} + B\overline{U}$ (A: $n \times n$ matrix; B: $n \times m$ matrix; $\overline{X}$: n-vector)

Note : because $\{X_e, U_e\}$ is an equilibrium, $\frac{dX_e}{dt} = \dot{X}_e = 0$ and $\frac{d\overline{X}}{dt} = \frac{dX}{dt} = \dot{X}$

Each entry in the A and B matrices come from the first-order Taylor expansion of the function f:

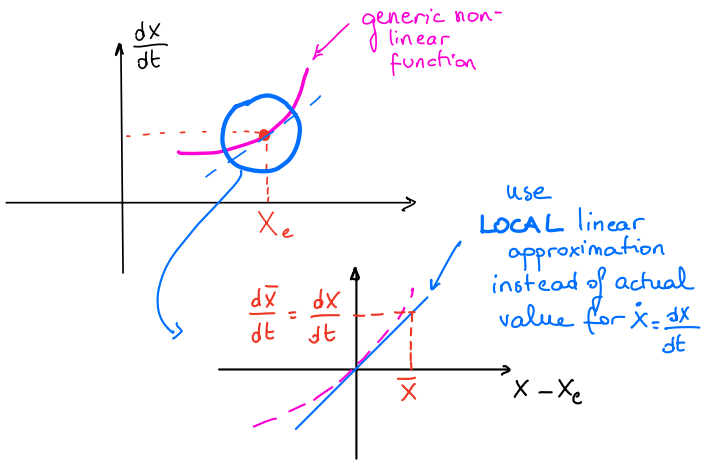
↳ only keep linear term

$$A = A_{ij} = \left.\frac{\partial f_i}{\partial X_j}\right|_{X_e, U_e}$$

i^{th} row $\equiv i^{th}$ governing equation; j^{th} column $\equiv j^{th}$ variable

$$B = B_{ij} = \left.\frac{\partial f_i}{\partial U_j}\right|_{X_e, U_e}$$

i^{th} row $\equiv i^{th}$ governing equation; j^{th} column $\equiv j^{th}$ control



What this looks like for longitudinal motion

- Variables of interest

$$X = \begin{pmatrix} V \\ \gamma \\ \alpha \\ q \\ h \end{pmatrix} \quad X_e = \begin{pmatrix} V_e \\ \gamma_e \\ \alpha_e \\ q_e = 0 \\ h_e \end{pmatrix} \quad \bar{X} = \begin{pmatrix} V - V_e \\ \gamma - \gamma_e \\ \alpha - \alpha_e \\ q \\ h - h_e \end{pmatrix} = \begin{pmatrix} \bar{V} \\ \bar{\gamma} \\ \bar{\alpha} \\ q \\ \bar{h} \end{pmatrix} \qquad f = \begin{pmatrix} f_{\boxed{P}} \\ f_{\boxed{S}} \\ f_{\boxed{Kq}} \\ f_{\boxed{D}} \\ f_{\boxed{Kh}} \end{pmatrix}$$

RHS of governing eqn.

- A and B matrices

state variables: $\bar{V}$ $\bar{\gamma}$ $\bar{\alpha}$ q $\bar{h}$

$$A = \begin{matrix} \\ \boxed{P} \\ \boxed{S} \\ \boxed{Kq} \\ \boxed{M} \\ \boxed{Kh} \end{matrix} \begin{bmatrix} A_{\boxed{P},V} & A_{\boxed{P},\gamma} & A_{\boxed{P},\alpha} & \cdots \\ A_{\boxed{S},V} & & & \\ A_{\boxed{Kq},V} & & & \\ \vdots & & & \end{bmatrix} \qquad B = \begin{bmatrix} B_{\boxed{P},\delta_x} & B_{\boxed{P},\delta_m} \\ B_{\boxed{S},\delta_x} & B_{\boxed{S}\,\delta_m} \\ \vdots & \vdots \end{bmatrix}$$

Equations

A: 5×5 matrix

B: 5×2

- **Example.** Linearization of the propulsion equation $\boxed{\mathbf{P}}$

Calculation of $A_{\boxed{P},\gamma}$

$$\boxed{P} \quad \frac{dV}{dt} = \overbrace{\frac{1}{m}\left(F\cos\alpha - D - mg\sin\gamma\right)}^{f_{\boxed{P}}(x,u)}$$

not explicit functions of γ

$$A_{\boxed{P},\gamma} = \frac{\partial}{\partial\gamma}\left(\frac{1}{m}\left(F\cos\alpha - D - mg\sin\gamma\right)\Big|_e\right)$$

$$= \frac{1}{m}\frac{\partial}{\partial\gamma}\left(-mg\sin\gamma\right)\Big|_e$$

$$A_{\boxed{P},\gamma} = \frac{1}{m}\left(-mg\cos\gamma\right)\Big|_e = -g\cos\gamma_e = -g$$

if starting from level flight

Calculation of $A_{P,V}$

$$A_{P,V} = \frac{\partial}{\partial V}\left(\frac{1}{m}\left(F\cos\alpha - D - mg\sin\gamma\right)\right)\Big|_e$$

need to refine how F depends on V

thrust model : $F = k_f \rho V^{\lambda_f} \delta_x$

λ_f function of the engine

$-1 < \lambda_f < 0$ for turbofan

- -1 propeller
- 0 simple flux engine
- 1 simple flux + afterburners
- 2 (sc) ramjets

* $$\frac{\partial F\cos\alpha}{\partial V}\Big|_e = \cos\alpha \frac{\partial F}{\partial V}\Big|_e$$

$$= \cos\alpha \frac{\partial}{\partial V}\left(k_f \rho V^{\lambda_f} \delta_x\right)\Big|_e$$

$$= \cos\alpha \left(k_f \rho \lambda_f V^{\lambda_f - 1} \delta_x\right)\Big|_e = \cos\alpha \underbrace{\left(k_f \rho V^{\lambda_f} \delta_x\right)}_{F} \frac{\lambda_f}{V}\Big|_e$$

$$= \cos\alpha_e F_e \frac{\lambda_f}{V_e}$$

* $$\frac{\partial D}{\partial V}\Big|_e = \frac{\partial}{\partial V}\left(\frac{1}{2}\rho S V^2 C_x\right)\Big|_e = \rho S V C_x\Big|_e = \rho S V_e C_x = 2\frac{D_e}{V_e}$$

* $$\frac{\partial(-mg\sin\gamma)}{\partial V}\Big|_e = 0$$

$$A_{P,V} = \frac{1}{m}\left[\underbrace{\cos\alpha_e}_{\approx 1} \lambda_f \frac{F_e}{V_e} - 2\frac{D_e}{V_e}\right]$$

Further simplification: α_e small, $F_e = D_e$ at equilibrium, $D_e = mg$

$f_R = \frac{C_2}{C_x}$

$$\boxed{A_{P,V} = \frac{g(\lambda_f - 2)}{f V_e}}$$

2 Mode analysis

The natural modes of the aircraft are determined by looking at the eigenvalues of the A matrix. Those modes will give us information about the dynamic stability of the aircraft:

> Natural modes of aircraft system
> = root λ_i of the characteristic polynomial of A
> $\mathrm{Det}(A - \lambda I) = 0$
> = eigenvalues of A

Each mode will be studied individually after an additional decoupling stage, and we'll be able to identify constraints on frequencies, damping, or characteristic times of each of them. Note that if the aircraft is naturally stable, it is "protected" against external perturbations. However, if the pilot acts on the controls, the modes will be modified! We need to make sure that those changes do not affect the overall stability of the aircraft.

Let's first review some general characteristics of modes. The temporal response to a given set of initial conditions is written:

change compared to equilibrium

$$\overline{X}_j = \sum_{i=1}^{n} a_i^j e^{\lambda_i t}$$

real or complex

If the root λ_i is complex, it is associated with its complex conjugate:

$$\lambda_i = u_i \pm i v_i$$

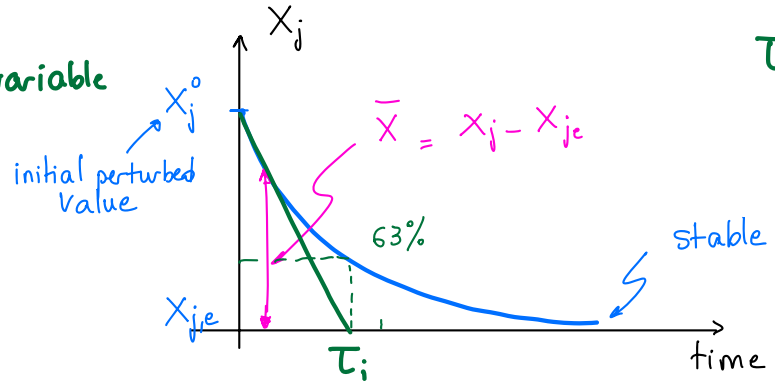
and the temporal response takes the form:

$$\overline{X}_j(t) = \sum_{i=1}^{n} a_i^j e^{v_i t} \sin\left(v_i t + \phi_i^j\right)$$

The A matrix is a square matrix of dimension $n \times n$: the system has n modes, which can be aperiodic or oscillatory.

Aperiodic modes. The eigenvalues for those modes are real ($v_i = 0$) and they are characterized by their time scale, which represents the time needed to reach 63% of its equilibrium value:

i^{th} mode
j^{th} state variable

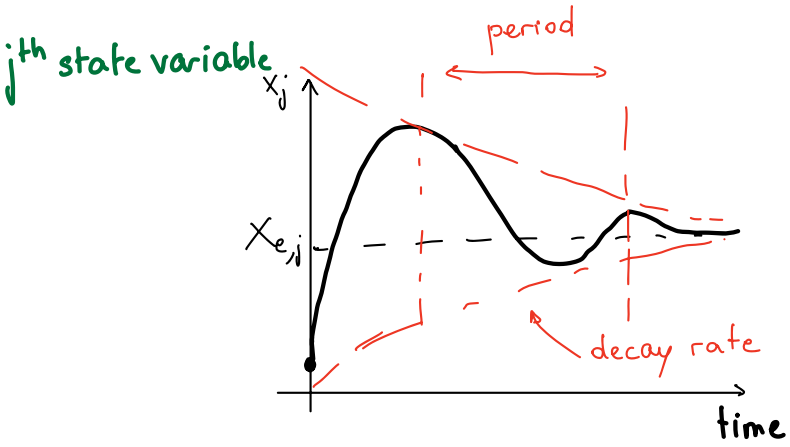


$$\tau_i = -\frac{1}{\lambda_i} = -\frac{1}{u_i}$$

<0 if stable for aperiodic

stable $\Rightarrow$ recover equilibrium value after some time

Oscillatory modes. The intrinsic frequency, also called damped frequency, ω_d is the real frequency of the mode, that directly measured on temporal responses. Expressed in rad/s, it is equal to the imaginary part of the eigenvalue:



frequency $\omega_d = \upsilon_i$ rad/s

damped

The period, measured in seconds, is equal to:

$$T = \frac{2\pi}{\omega_d}$$

The undamped, or natural frequency, ω_n of the mode is obtained from the damped one through the relationship:

$$\omega_n = \frac{\omega_d}{\sqrt{1-\zeta^2}}$$

where ζ is the damping ratio. ζ is equal to the real part of the eigenvalue:

$$u_i = -\zeta\omega_n$$

$$\zeta = -\frac{u_i}{\omega_n} = \frac{1}{\sqrt{1+(\upsilon_i/u_i)^2}}$$

ζ is responsible for the oscillatory nature of the response, for example, how many oscillations will be visible. $\zeta\omega_n$ is equivalent to the inverse of a time scale, which is characteristic of the envelope of the response. From those parameters, we can write a characteristic 2nd degree polynomial of the response:

2 x 2 matrix

$\Rightarrow$ 2nd order characteristic polynomial

$$\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2 = 0$$

$\lambda \equiv$ pair of complex conjugate

Application to longitudinal motion Simplifying somewhat the notations, the full linearized system for the longitudinal equations can be written:

- [P] propulsion equation
- [S] sustentation
- [Kq] kinematic equation for pitch velocity
- [M] moment
- [Kh] kinematic equation for altitude

$$X = \begin{pmatrix} V \\ \gamma \\ \alpha \\ q \\ h \end{pmatrix} \quad X_e = \begin{pmatrix} V_e \\ \gamma_e \\ \alpha_e \\ q_e \\ h_e \end{pmatrix}$$

$\gamma_e = 0$ often

$q_e = 0$ always

Linearized system of equations

$$\begin{pmatrix} \dot{\bar{V}} \\ \dot{\bar{\gamma}} \\ \dot{\bar{\alpha}} \\ \dot{q} \\ \dot{\bar{h}} \end{pmatrix} = \begin{matrix} P \\ S \\ K_q \\ M \\ K_h \end{matrix} \begin{bmatrix} A_V & A_\gamma & A_\alpha & A_q & A_h \\ B_V & 0 & B_\alpha & B_q & B_h \\ C_V & 0 & C_\alpha & C_q & C_h \\ D_V & 0 & D_\alpha & D_q & D_h \\ 0 & E_\gamma & 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} \bar{V} \\ \bar{\gamma} \\ \bar{\alpha} \\ q \\ \bar{h} \end{pmatrix} + \begin{bmatrix} A_{\delta_x} & A_{\delta_m} \\ B_{\delta_x} & B_{\delta_m} \\ C_{\delta_x} & C_{\delta_m} \\ D_{\delta_x} & D_{\delta_m} \\ E_{\delta_x} & E_{\delta_m} \end{bmatrix} \begin{pmatrix} \bar{\delta}_x \\ \bar{\delta}_m \end{pmatrix}$$

matrix for long period mode

matrix for short period mode

$A \equiv 5 \times 5$ matrix $\Rightarrow$ 5 eigenvalues $\begin{cases} 4 \text{ complex} \\ 1 \text{ real} \end{cases} \Rightarrow \begin{cases} 2 \text{ oscillatory modes} \\ 1 \text{ aperiodic mode} \end{cases}$

- Mode 1 oscillatory : associated with V and γ
  - ↳ long period mode, phugoid
- Mode 2 oscillatory : associated with q and α
  - ↳ short period mode
- Mode 3 : associated with altitude h
  - ↳ propulsion feedback mode

ANALYTICAL

Lecture notes, homework

"Can" handle 2nd order characteristic polynomials (2×2 matrix)
- ↳ requires decoupling assumptions
- ↳ allows physical interpretations
- ↳ lower accuracy

NUMERICAL

Case studies

Full temporal response of aircraft
- ↳ more accurate
- ↳ more difficult to extract physics

3 Phugoid or long-period mode

This is the first mode we'll study in detail because it can be easily analyzed "by hand". From the piloting/flight dynamics point of view, it is not problematic because it is a slow mode, nearly always stable and well-damped. It can be handled easily by both human- and auto-pilot. It is also historically the first mode to have been discovered and studied by Lanchester in the late 1800s. To facilitate the analysis, we need at most a 2×2 matrix, in this case, we want to decouple the A matrix in V and γ, which can be done if

Assumption : terms associated with
A_α, A_q, A_h } are 0 or negligible
B_α, B_q, B_h }

This decoupling assumption will be verified *a posteriori* with numerical results.

3.1 Initial physical considerations

Long-period mode is an oscillatory mode between V and γ, happening at a constant angle of attack

3.2 Decoupled equations

looking at natural mode only: no control

After decoupling the equations (matrix A), and adding the contributions from the controls, we get: with

$$\bar{V} = V - V_e \qquad \bar{\gamma} = \gamma - \gamma_e \quad (=0)$$

$$\begin{pmatrix} \dot{\bar{V}} \\ \dot{\bar{\gamma}} \end{pmatrix} = \frac{d}{dt}\begin{pmatrix} \bar{V} \\ \bar{\gamma} \end{pmatrix} = \frac{d}{dt}\begin{pmatrix} V - V_e \\ \gamma - \gamma_e \end{pmatrix} = \begin{pmatrix} \dot{V} \\ \dot{\gamma} \end{pmatrix} = \begin{bmatrix} A_V & A_\gamma \\ B_V & 0 \end{bmatrix} \begin{bmatrix} \bar{V} \\ \bar{\gamma} \end{bmatrix}$$

with
$$\begin{cases} A_V = \dfrac{g(\lambda_f - 2)}{f\, V_e} \\ A_\gamma = g \\ B_V = \dfrac{2g}{V_e^2} \end{cases}$$

λ_f = lift to drag ratio

The equilibrium velocity V_e is the velocity around which the equations have been linearized. Let's look more closely at the linearization process to extract some physical understanding of the frequency and damping terms:

- Propulsion equation **P** :

$$m\dot{V} = F - D - mg\gamma$$

$D = \frac{1}{2}\rho S V^2 C_x$

- Sustentation (Lift) equation **S** :

$$mV\dot{\gamma} = L - mg$$

$L = \frac{1}{2}\rho S V^2 C_z$

- The thrust balance changes with velocity V:

$$\frac{\partial (F - D)}{\partial V} = \frac{\partial F}{\partial V} - \frac{\partial D}{\partial V} = F_V - D_V = mg\frac{\lambda_f - 2}{f\, V_e}$$

The terms F_V and D_V are the gradients of thrust and drag with respect to the velocity. The linearized equations (consistent with A above) therefore are:

[P] $$m\dot{\bar{V}} = (F_V - D_V)\bar{V} - mg\bar{\gamma}$$

[S] $$m V_e \dot{\bar{\gamma}} = \frac{2mg}{V_e}\bar{V}$$

Comments:

$$\dot{\bar{\gamma}} = \frac{2g}{V_e^2}\bar{V} \Rightarrow$$ can interpret $\bar{\gamma}$ as being the integral of $\bar{V}$ (with some coefficients involved)

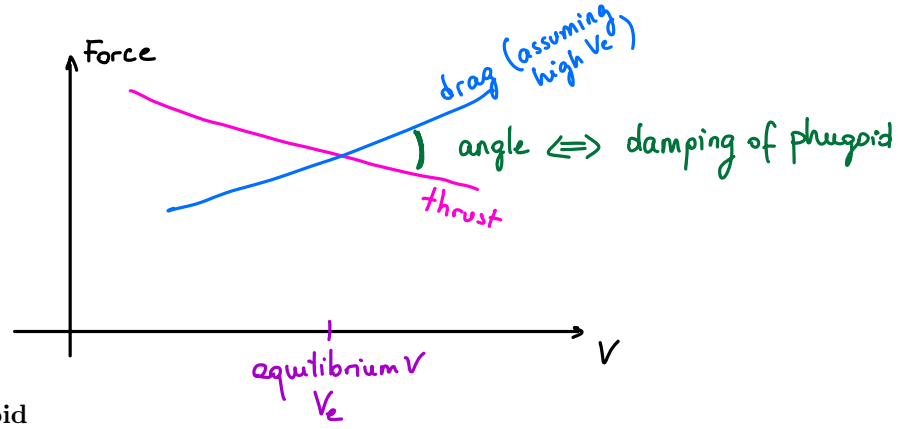
$\frac{2g}{V_e^2}$: constant

restoring force

$$m\dot{\bar{V}} = (F_V - D_V)\bar{V} - mg\bar{\gamma}$$

$\dot{\bar{V}}$: $\ddot{x}$ — $(F_V - D_V)$: damping term — $\bar{V}$: $\dot{x}$ — $\bar{\gamma}$: x

Harmonic oscillator interpretation



Undamped phugoid

damping term $= 0 = F_V - D_V$ (regardless of velocity)

linearized [P] $$m\dot{\bar{V}} = -mg\bar{\gamma} \Rightarrow \gamma = -\frac{\dot{V}}{g}$$

($\gamma_e = 0$: $\bar{\gamma} = \gamma$)

linearized [kh] $$\dot{\bar{h}} = \bar{V}\gamma$$

$$\Rightarrow \dot{\bar{h}} = \bar{V}\left(-\frac{\dot{V}}{g}\right) = -\frac{1}{g}\bar{V}\dot{V}$$

integrate: $$h = -\frac{V^2}{2g} + \boxed{\text{constant}}$$ (account for the equilibrium variables)

or $$\boxed{h + \frac{V^2}{2g} = \text{constant}} \Rightarrow$$ equation for "total" altitude

$\Rightarrow$ parallel with potential energy AND kinetic energy

3.3 Solving the equations

The characteristic polynomial for the phugoid can be used to find the eigenvalues, and therefore, the modes. Since the decoupled matrix is triangular, this is easily done:

$$\begin{pmatrix} \dot{\bar{V}} \\ \dot{\bar{\gamma}} \end{pmatrix} = \underbrace{\begin{bmatrix} A_V & A_\gamma \\ B_V & 0 \end{bmatrix}}_{\underline{\underline{A}}} \begin{pmatrix} \bar{V} \\ \bar{\gamma} \end{pmatrix}$$

Eigenvalues: $\det\left(\underline{\underline{A}} - \lambda \underline{\underline{I}}\right) = 0$

$$\begin{vmatrix} A_V - \lambda & A_\gamma \\ B_V & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow (A_V - \lambda)(-\lambda) - B_V A_\gamma = 0$$

$$\Rightarrow \lambda^2 - A_V \lambda - A_\gamma B_V = 0$$

Characteristic polynomial: $\lambda^2 + 2\zeta\omega_n \lambda + \omega_n^2 = 0$

- **Frequency** (natural)

$$\omega_n = \sqrt{-A_\gamma B_V} = \sqrt{2}\,\frac{g}{V_e}$$ (see expressions for matrix elements above)

- **Period** (neglect damping $\Rightarrow \omega_d \approx \omega_n$)

$$T = \frac{2\pi}{\omega_d} \simeq \frac{2\pi}{\omega_n} = \sqrt{2}\,\frac{\pi}{g} V_e \approx 0.45\, V_e$$

↑ Everything in SI!

- **Damping ratio**

$$\zeta = -\frac{A_V}{2\omega_n} = \frac{2 - \lambda_f}{2\sqrt{2}\, f}$$ ← lift to drag ratio

Comments

* Period is very simple! $\approx \frac{1}{2} V_e$ IN SECONDS (and V_e in m/s)
  - ↳ does not depend on aircraft
  - ↳ period is very long (hence the name!)
* Damping occurs when $2\zeta\omega_n \geq 0$ $\Rightarrow$ ζ depends on λ_f, so on the type of engine

$$\lambda_f = \begin{cases} -1 & \text{propeller} \\ 0 & \text{single flux engine} \\ \approx 2 & \text{for scramjet/ramjet} \end{cases}$$

$-1 \leq \lambda_f \leq 0$ for turbofan

ζ always ≥ 0 except when dealing with supersonic

BUT assumed $\gamma_e = 0$ in analysis.

3.4 Phugoid instability

The phugoid mode can become unstable, for example for fighter jets flying with a high slope:

This result can be obtained by linearizing around a non-zero slope. In this case, the instability appears for:

$$\zeta\big|_{\gamma_e \neq 0} = \frac{2-\lambda_f}{f(1+\lambda_f)} - \tan\gamma_e$$

$$\text{Conditions for } \zeta \text{ to become } <0 \;:\; \tan\gamma_e \geq \frac{2-\lambda_f}{f(1+\lambda_f)}$$

- Propeller or double-flux (bypass) engine:

  $\lambda_f \approx -1$ typical for commercial aircrafts

  ↳ denominator goes to ∞ : only option for $\zeta<0$ is $\gamma_e \approx 90°$

  ↳ CANNOT BE UNSTABLE

- Simple flux engine:

  fighter jets : $\lambda_f \approx 0$

  $\tan\gamma_e \geq \frac{2}{f} \approx {}^{2}/_{10} = 0.2 \rightarrow 12°$

  $f \leftarrow \approx 10$ (low side)

  CAN BECOME UNSTABLE AT HIGH CLIMBING SLOPES

- Ramjet:

  $\lambda_f \approx 2 \Rightarrow$ ALWAYS UNSTABLE

4 Propulsion feedback mode [PFM]

This is a very slow, aperiodic mode that brings the slope back to zero. The characteristic time scale is linked to the change in thrust due to changes in altitude, constrained by:

* α = constant
* m = constant] PFM operates over a long time scale, but small enough that fuel consumption is negligible.

4.1 Physical considerations (with support from the equations)

We assume all equations at equilibrium except for the altitude kinematic equation [Kh] :

$\dot{h} = V\gamma \Rightarrow$ what we are looking at: influence of the change in **density** on quasi-steady flight equation

We assume again that the angle of attack does not vary:

- [S] : α = const, m = const $\Rightarrow$ C_z = constant $\Rightarrow$ ρV^2 = constant; C_x = constant $\Rightarrow D = \frac{1}{2}\rho V^2 C_x$ = constant
- [P] $F - D = mg\gamma \Rightarrow \Delta F = mg\,\Delta\gamma$ (D: constant) — any change in thrust linked directly change in slope.
- From the thrust model: $F = k\rho V^{\lambda_f}\delta_x = k_f a V^{\lambda_f - 2}\delta_x$ (replace ρ) (1) (k: constant; k_f: constant; ρV^2 = constant from [S]) or $= k_f a^{\lambda_f/2}\rho^{1-\lambda_f/2}\delta_x$ (replace V) (2)

So when altitude increases,

(2) $\lambda_f \leq 2 \Rightarrow$ if $h\uparrow$, $\rho\downarrow$, exponent $\geq 0 \Rightarrow F\downarrow$

(1) ρV^2 = constant $\Rightarrow$ if $h\uparrow$, $\rho\downarrow$, $V\uparrow$, exponent < 0, $\Rightarrow F\downarrow$

if $h\uparrow$, $F\downarrow \Rightarrow \Delta F < 0 \Rightarrow \Delta\gamma < 0$ slope tends to 0, climb levels out (if keep δ_x constant)

4.2 Equations

We are interested in estimating the characteristic time associated with this mode. To do that, we'll consider the phugoid system, and add **Kh** to it. We assume the following:

- After any change in angle of attack (always fast), the angle of attack stays constant.
- Pitch velocity nearly zero.
- Therefore, we are in a pseudo-equilibrium situation where only h and γ vary.

$$\begin{matrix}\text{P}\\ \text{S}\\ \text{Kh}\end{matrix}\begin{pmatrix}\dot{\bar{V}}\\ \dot{\bar{\gamma}}\\ \dot{\bar{h}}\end{pmatrix} = \begin{bmatrix} A_V & A_\gamma & A_h \\ B_V & 0 & B_h \\ \boxed{0} & V_e & 0 \end{bmatrix}\begin{pmatrix}\bar{V}\\ \bar{\gamma}\\ \bar{h}\end{pmatrix}$$

$\bar{V}$ $\bar{\gamma}$ $\bar{h}$

$$\frac{\partial}{\partial V}\left(V\sin\gamma\right)\Big|_e = \sin\gamma_e$$

γ_e small or 0

The smallest root of the characteristic polynomial (3rd degree here) is the one we are looking for.

3×3 matrix $\rightarrow$ 3rd degree polynomial : hard to solve analytically

but τ expected to be large $= -\frac{1}{\lambda_{PFM}}$, so $|\lambda_{PFM}|$ small !

↳ $|\lambda_{PFM}|^3 \approx |\lambda_{PFM}|^2$

↳ $\lambda_{PFM} = \dfrac{V_e B_h A_V}{V_e B_h + A_\gamma B_V}$ $\Rightarrow$ indeed, real, ≤ 0

aperiodic mode

$$\tau_{PFM} = -\frac{1}{\lambda_{PFM}} = \frac{f}{2-\lambda_f}\left(\frac{V_e}{g} + 66\,\frac{T_e}{V_e}\right)$$

temperature at equilibrium, appears because of B_h

DIMENSIONAL NUMBER ! CORRECT ONLY IF USING SI

5 Short-period mode (angle of attack oscillatory mode)

This mode is the most important one for longitudinal flight dynamics qualities:

- Occurs at frequencies slightly higher than "pilot" frequencies (≈ 1 Hz)
  - ↳ difficult to control manually
- Can become unstable
  - ↳ if non-optimal location of G with respect to aero center
  - ↳ if bad coupling with pilot reaction

We saw earlier, while looking at longitudinal equilibrium, one approximate criterion for its stability: *static* stability.

"G in front of A"
$\Rightarrow$ necessary, but not sufficient

We will see however that there are consequences of having G *too much* in front of A. As above, we need a 2×2 matrix for proper analytical study, isolating in this case α and q. To do so, we formulate the following assumptions:

Assumptions : velocity and altitude changes SLOW compared to changes in α & q
V & h ≈ constant over the time scale associated with SPM.

5.1 Physical considerations

This mode is a pitch oscillation around a point located *in front of* the center of gravity of the aircraft. The feedback moment is created by the change in lift applied at A, behind the center of gravity:

$$C_{m_\alpha} < 0$$

and this moment will determine the frequency. Damping is created by the gradient of the pitching moment with respect to the pitch velocity:

$$C_{m_q} = \frac{\partial C_m}{\partial q}$$

and the gradient in lift coefficient with respect to the angle of attack:

$$C_{z_\alpha} = \frac{\partial C_z}{\partial \alpha}$$

5.2 Decoupled equations

According to our decoupling hypotheses, and accounting for controls, we obtain the following system:

$$\bar{\alpha} = \alpha - \alpha_e$$
$$\bar{q} = q - \underbrace{q_e}_{0} = q$$

$$[K_q] \begin{pmatrix} \dot{\bar{\alpha}} \\ \dot{q} \end{pmatrix} = \underbrace{\begin{pmatrix} C_\alpha & C_q \\ D_\alpha & D_q \end{pmatrix}}_{\underline{\underline{A}}\ \text{SPM}} \begin{pmatrix} \bar{\alpha} \\ q \end{pmatrix}$$

$[M]$

A_{SPM} matrix:

[Kq] equation: $\dot{\alpha} + \dot{\gamma} = q \Rightarrow \dot{\alpha} = q - \dot{\gamma}$

↳ take $\dot{\gamma}$ from [S]: $\dot{\gamma} = \frac{1}{2}\frac{\rho S V^2 C_z}{mV} - \frac{mg}{mV}$

$$\Rightarrow \dot{\alpha} = \underbrace{q - \frac{1}{2}\frac{\rho S V^2 C_z}{mV} + \frac{mg}{mV}}_{\text{RHS}}$$

* $C_\alpha = \frac{\partial RHS}{\partial \alpha} = -\frac{1}{mV}\frac{1}{2}\rho S V^2 C_{z_\alpha} \quad (= -B_\alpha)$

* $C_q = \frac{\partial RHS}{\partial q} = 1 - \frac{\partial}{\partial q}\left(\frac{1}{2}\frac{\rho S V^2 C_z}{mV}\right)$

$= 1 - \frac{1}{2}\frac{\rho S V^2}{mV}\boxed{\frac{\partial C_z}{\partial q}}$, $\boxed{\frac{\partial C_z}{\partial q} \approx 0}$

$C_q = 1 \quad (= 1 - B_q)$

[M] equation: $D_\alpha = \frac{1}{B}\left(\frac{1}{2}\rho S \ell V^2 C_{m_\alpha}\right)$

$D_q = \frac{\ell}{BV_e}\left(\frac{1}{2}\rho S \ell V^2 C_{m_q}\right)$

What is B? products of inertia

$$I = \begin{pmatrix} A & D & E \\ D & \boxed{B} & F \\ E & F & C \end{pmatrix} \rightarrow \int (x^2+z^2)\,dm$$

B ≡ moment of inertia with respect to y_b axis

moments of inertia.

Eigenvalues of matrix $A_{SPM} = \begin{pmatrix} -B_\alpha & 1 \\ D_\alpha & D_q \end{pmatrix}$ are the roots of:

$$\lambda^2 + \left(-D_q + B_\alpha\right)\lambda \boxed{- D_\alpha - D_q B_\alpha} = 0$$

compared to harmonic oscillator $\lambda^2 + 2\zeta\omega_n\lambda + \boxed{\omega_n^2} = 0$

Natural oscillation frequency:

By inspection: $\omega_n^2 = -D_\alpha - D_q B_\alpha$

$$\omega_n^2 = -\frac{1}{2}\rho S V^2 \frac{\ell}{B}\left[C_{m_\alpha} + \underbrace{\frac{1}{2m}\rho S \ell C_{m_q} C_{z_\alpha}}_{\ll C_{m_\alpha} \text{ if } |m_s| \text{ is not too small}}\right]$$

- if altitude increase: $\omega_n \downarrow$
- if static margin is large (in absolute value):

$$C_{m_\alpha} = \underbrace{\frac{x_A - x_G}{\ell}}_{\text{static margin.}} C_{z_\alpha} \Rightarrow \text{large } |m_s| \rightarrow \text{large } |C_{m_\alpha}| \rightarrow \text{high } \omega_n$$

- if velocity increases: $\omega_n \uparrow$
- if aircraft flying at high α: need to rewrite moment of inertia B

$B = m R^2$ (R: radius of gyration)

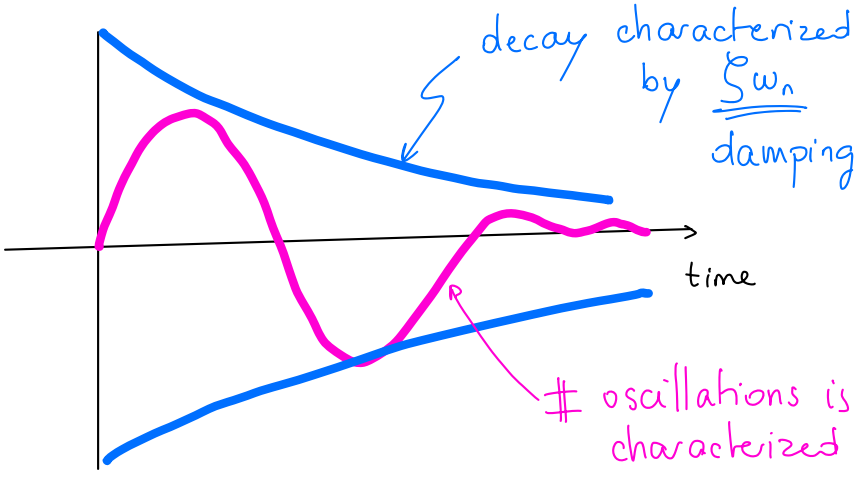
... math ...

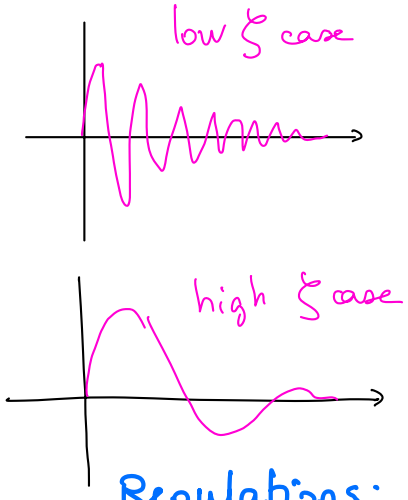
alternative expression: $\omega_n = \frac{1}{R}\sqrt{g(x_G - x_A)}\sqrt{\frac{C_{z_\alpha}}{C_z}}$

ω_n does depend on α

if α / ℓ increases, $\omega_n \downarrow$

Damping characteristics





Regulations: $\zeta > 0.4$

$$\zeta\omega_n = \frac{1}{2V_e}\frac{1}{2}\rho_e S V_e^2\left(\underbrace{-\frac{\ell^2}{B}C_{m_q}}_{①} + \underbrace{\frac{C_{z_\alpha}}{m}}_{②}\right)$$

① ≈ ②

Each term contributes roughly equally to $\zeta\omega_n$

Simple model for C_{m_q}

$$C_{m_q} = -\left(\frac{S_E}{S}\right)\left(\frac{d}{\ell}\right)^2 C_{z_{\alpha_E}}$$

← moment created by q depends on the change of lift force generated by the empennage

Damping does not depend on static margin

Damping ratio :

$$\zeta = \frac{1}{\sqrt{-C_{m_\alpha}}}\sqrt{\frac{\rho S B}{8\ell}}\left(-\frac{\ell^2}{B}C_{m_q} + \frac{C_{z_\alpha}}{m}\right)$$

- ζ independant of velocity
- $\zeta \downarrow$ when $h \uparrow$ (more oscillations at high altitude)
- ζ depends on static margin
  - if G moves forward : $|C_{m_\alpha}| \uparrow \rightarrow \zeta \downarrow$: more oscillations
  - may lead to pilot induced / pilot-augmented oscillations
  - may lead to destabilization of aircraft

Note : if G moves backward ($|m_s|$ decreases), the oscillatory modes transition to 2 aperiodic modes, but one of them is unstable (until G ends up behind F, in which case the aircraf if statically unstable)